

# AsliChoice — Project Details

Source: `SOFTWARE REQUIREMENT SPECIFICATION ASLI CHOICE.docx` This file is a structured, planning-ready summary of the SRS. Use it as the working reference for module design, schema planning, and phasing.

## 1. Platform Overview

**AsliChoice** is an omnichannel commerce + operations platform covering:

- Retail POS (offline-capable)
- Online Store (location-aware ecommerce)
- Inventory across branches & warehouses
- Procurement (centralized) and Vendor management
- CRM, Referral & Wallet
- Finance, Accounting & GST
- HR, Attendance & Payroll
- Reporting / BI
- Logistics & Fulfillment
- Returns / QC
- System configuration, Security, Audit
- Mobile apps + Offline sync + APIs

### Cross-cutting Architectural Principles

| Principle                                            | Meaning                                                                                          |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Centralized Inventory + Distributed Branch Ownership | One inventory engine; stock belongs to Branch + Warehouse.                                       |
| Ledger-Driven                                        | Inventory, wallet, and finance never allow direct edits — only ledger entries.                   |
| Branch-Aware                                         | POS, fulfillment, payments, reports, security all scope by branch.                               |
| Pincode-Aware                                        | Online inventory visibility, MOV, delivery, serviceability, fulfillment all depend on pincode.   |
| Omnichannel Unified Engine                           | POS + Online + future Wholesale share inventory, customers, offers, wallet, referral, reporting. |
| Negative Stock — Strictly Prohibited                 | Hard-block, not warning. Inventory can hit 0, never below.                                       |
| Configurable, Not Hardcoded                          | Global + branch-override settings; feature toggles.                                              |
| API-First, Offline-Capable                           | Mobile apps, POS, attendance must work offline and sync later.                                   |
| Audit-Everywhere                                     | Every critical action creates an immutable audit log.                                            |
| Non-MLM Referral                                     | Single-level only; successful referrer auto-becomes Partner.                                     |
| Soft Delete Only for masters                         | Never permanently delete masters; <code>is_active = false</code> .                               |

## 2. Module Index

| #  | Module                                      | Role                                                                            |
|----|---------------------------------------------|---------------------------------------------------------------------------------|
| 1  | Master Management                           | Single source of truth (products, pricing, branches, taxes, customers, vendors) |
| 2  | User & Role Management                      | Auth, roles, permissions, sessions, audit                                       |
| 3  | Vendor Management                           | Procurement-ready supplier ecosystem                                            |
| 4  | Purchase Management                         | Centralized procurement + branch-aware GRN                                      |
| 5  | Inventory Management                        | Ledger-driven, branch + warehouse stock                                         |
| 6  | Online Store                                | Location-aware ecommerce                                                        |
| 7  | Retail POS                                  | Branch-scoped, offline-capable billing                                          |
| 8  | Sales & Order Management                    | Unified commerce transaction engine                                             |
| 9  | CRM & Customer Management                   | Central customer intelligence                                                   |
| 10 | Referral, Partner & Wallet                  | Non-MLM growth & incentive engine                                               |
| 11 | Notifications & Communication               | OTP/SMS/Email/WhatsApp/Push                                                     |
| 12 | Finance, Accounts & Ledger                  | Branch-aware accounting & GST                                                   |
| 13 | Reports, Analytics & BI                     | Role-based dashboards & exports                                                 |
| 14 | Delivery, Logistics & Fulfillment           | Pincode + branch-based fulfillment                                              |
| 15 | Returns, Damages & QC                       | Inventory recovery & quality assurance                                          |
| 16 | HR, Staff, Attendance & Payroll             | Phase 1 includes payroll                                                        |
| 17 | Document, Media & File Management           | Central digital asset engine                                                    |
| 18 | System Settings, Configuration & Automation | Global + branch configs, toggles, cron, workflows                               |
| 19 | Super Admin, Security & Audit               | Enterprise governance                                                           |
| 20 | Mobile App, Offline Sync & API              | Mobility + digital connectivity                                                 |

## MODULE 1 — Master Management

**Foundation. Single source of truth for products, pricing, branches, taxes, customers, vendors. All other modules depend on it.**

### Submodules

1. **Product Master** — Name, SKU, Barcode, HSN, Brand, Images, Active/Online/POS flags. Types: Simple, Variant, Combo, Service.

2. **Category Management** — Multi-level parent-child.
3. **Product Variants** — Separate SKU/barcode/inventory/price per variant (size, flavor, weight, packaging).
4. **Unit Management** — Kg, g, L, Piece, Bottle; supports conversions (1 Box = 12 Bottles).
5. **Brand Management** — Name, logo, description, status.
6. **Tax Management (GST)** — CGST/SGST/IGST, inclusive/exclusive, product-wise.
7. **Price Management** — Categories: MRP, Retail, Wholesale, Partner, Internal Consumption. Branch-wise ready.
8. **Internal Consumption Pricing** — Super Admin only; no backdated entries; mandatory comments; immediate inventory reduction; separate reporting.
9. **Branch & Warehouse Management** — Multiple branches/warehouses, branch-wise stock, GST/contact info.
10. **Customer Master** — Name, mobile, email, address, optional GST.
11. **Vendor Master** — Vendor name, contact, GST, payment terms.
12. **Payment Mode Master** — Cash, UPI, Card, Wallet, Bank Transfer.
13. **Offer/Coupon Master** — %/flat/product-based, expiry.

## Business Rules

- Structured SKU (e.g. GHE-GIR-500 ).
- Barcode-ready from Day 1.
- Soft delete only.
- Multi-branch ready even if launching single-branch.
- Audit logs for product/price/inventory-affecting changes.

## Core Tables

products, product\_variants, categories, brands, units, taxes, price\_lists, branches, warehouses, customers, vendors, payment\_modes, offers, internal\_consumptions

---

# MODULE 2 — User & Role Management

**Security backbone. Auth, authorization, roles, permissions, branch access, sessions, audit, user hierarchy.**

## Architecture

- **Single unified user table** for customers, partners, staff, admins, vendors.
- One user can have multiple roles (e.g. Customer + Partner).
- Structure: User → Roles → Permissions → Branch Access → Profiles .

## Authentication

- **Identifiers:** Mobile, Email.
- **Methods:** OTP login, Password login.
- **OTP Channels:** SMS, Email.
- **Smart OTP Fallback:** If user enters mobile but mobile OTP is disabled and email OTP is enabled with verified email → send to email. Reverse also supported. *User input ≠ OTP destination.*
- **Super Admin controls:** enable/disable each identifier, method, channel; OTP length, expiry, cooldown, max attempts.
- **Validation guard:** System blocks invalid combos (e.g. Mobile login on + Mobile OTP off + Password off).

## Roles

SUPER\_ADMIN, ADMIN, MANAGER, STAFF, CASHIER, CUSTOMER, PARTNER, VENDOR

## Permissions

Granular VIEW / CREATE / EDIT / DELETE / APPROVE / EXPORT per resource (e.g. PRODUCT\_VIEW , PRODUCT\_CREATE ).

## Branch Access

Single / Multi / All-branch.

## Profiles

Common (name, mobile, email, address, DOB, photo) + Specialized (Staff: empld/dept; Partner: referral code/wallet; Vendor: GST/company).

## Referral → Partner Flow

Customer refers → referred customer completes successful purchase → original customer **auto-becomes Partner**.  
Single-level only. No MLM.

## Wallet

Commission → Internal Wallet. No payouts initially; payout-ready schema preserved.

## Security

JWT + refresh tokens, role + permission middleware, route guards, permission-based menus, session tracking, force logout, OTP rate limiting.

## Tables

users, roles, permissions, user\_roles, role\_permissions, user\_permissions, user\_branches, sessions, audit\_logs, partner\_profiles, staff\_profiles, vendor\_profiles, customer\_profiles

---

# MODULE 3 — Vendor Management

**Procurement foundation. Vendor ≠ contact record — a fully trackable procurement entity with transaction, pricing, payment, supply, and analytics history.**

## Submodules

1. **Vendor Master** — Business (Name, Company, Code, GST, PAN, Category, Business Type), Contact, Addresses (Billing/Pickup/Warehouse), Banking, Tax (GST type/state, future TDS). Vendor Code auto-generated (e.g. VEN-0001 ).
2. **Vendor Categories** — RAW\_MATERIAL, PACKAGING, TRANSPORT, SERVICE\_PROVIDER, DISTRIBUTOR, MANUFACTURER .
3. **Status** — ACTIVE, INACTIVE, BLACKLISTED, ON\_HOLD . Blacklisted cannot transact.
4. **Contact Persons** — Multiple per vendor (Accounts, Sales, Dispatch).
5. **Payment Terms** — Advance, Net 7/15/30; default + PO-level override.
6. **Pricing** — Vendor-specific rates; **price history with effective dates**.
7. **Vendor-Product Mapping** — Multiple vendors per product, preferred vendor, lead time, MOQ.
8. **Performance Tracking** — Delivery on-time, quality/rejection, pricing stability, volume; vendor scorecards (future).
9. **Documents** — GST cert, agreements, quotations, licenses; upload + expiry tracking.
10. **Communication Log** — Calls, emails, meetings, negotiations.

11. **Notes & Remarks** — Internal only, not vendor-visible.
12. **Vendor-Branch Mapping** — Optional, for local procurement.
13. **Approval Workflow** (future) — Created → Verification → Approval → Activated.
14. **Ledger** (future finance) — Total purchases, payments, outstanding, debit/credit notes. Finance-ready schema from Day 1.
15. **Vendor Portal** (future).

## Validations

- GST format validation mandatory.
- Duplicate detection on GST / mobile / company.
- Soft delete only.

## Permissions

VENDOR\_VIEW, VENDOR\_CREATE, VENDOR\_EDIT, VENDOR\_APPROVE, VENDOR\_EXPORT

## Tables

vendors, vendor\_contacts, vendor\_categories, vendor\_products, vendor\_price\_history, vendor\_documents, vendor\_notes, vendor\_payment\_terms, vendor\_communications, vendor\_branch\_mapping, vendor\_ledgers

# MODULE 4 — Purchase Management

**Centralized procurement + distributed inventory. Procurement controlled by HO/Main Branch; inventory belongs to branch + warehouse.**

## Procurement Scenarios

- **Scenario 1 — Central Warehouse:** Vendor → HO Purchase → HO Warehouse GRN → HO Inventory → Branch Transfer.
- **Scenario 2 — Direct Branch:** HO PO → Vendor delivers to Branch → Branch GRN → Branch Inventory.

## Submodules

1. **Purchase Requisition** — Internal request. Only HO/Main Branch creates. Multiple requisitions can merge into one PO. Doesn't affect inventory. Priorities: LOW/MEDIUM/HIGH/URGENT . Status flow: DRAFT → SUBMITTED → APPROVED → PARTIALLY\_PROCESSED → COMPLETED → REJECTED .
2. **Purchase Order** — Header (PO#, Vendor, Procurement Branch, **Destination Branch**, **Destination Warehouse**, dates, terms) + Items (product/variant/qty/unit/rate/tax/discount). Statuses: DRAFT → PENDING\_APPROVAL → APPROVED → PARTIALLY\_RECEIVED → COMPLETED / CANCELLED . Auto-generated PO numbers.
3. **Approval Workflow** — Small auto-approve, large needs admin. Future: Manager → Admin → Super Admin. **Approval thresholds are configurable** (per-amount, per-category, per-branch) rather than hardcoded.
4. **GRN (Most Critical)** — **Inventory updates ONLY after GRN**, never after PO. GRN updates Destination Branch + Warehouse inventory. Fields: GRN#, linked PO, dest branch/warehouse, received qty, damaged qty, receiver, notes.
5. **Partial Receipt** — Multiple inwarding under one PO.
6. **Purchase Invoice** — Invoice upload, GST, tax breakup, PO + GRN + vendor linkage. Finance-ready.
7. **Purchase Returns** — Reasons: DAMAGED, WRONG\_PRODUCT, EXPIRED, QUALITY\_ISSUE . Affects branch inventory, warehouse stock, vendor ledger, analytics.
8. **Vendor Credit & Payment** — Outstanding, due dates, advance, partial payments. Modes: Cash, UPI, Bank, Cheque.

9. **Attachments** — Invoices, quotations, transport receipts.
10. **Notes & Remarks** — Internal.
11. **Inventory Integration** — Movement type `PURCHASE` . Inventory belongs to Dest Branch + Warehouse.
12. **Multi-branch Procurement** — HO can purchase for any branch. **Branches cannot directly purchase from vendors.**
13. **Analytics** — Branch/vendor/product purchase trends, cost trends, delayed delivery, returns.
14. **Reports** — Purchase Register, PO, GRN, Branch/Vendor-wise, Pending PO, Returns, Outstanding Vendor.
15. **Barcode Integration** — Inward scanning, GRN verification.
16. **Offline Support** (future) — Queue-based sync for warehouses & low-internet branches; offline GRN capture, local queue, conflict-aware sync on reconnect.

## Permissions

`PURCHASE_VIEW`, `PURCHASE_CREATE`, `PURCHASE_APPROVE`, `PURCHASE_GRN`, `PURCHASE_RETURN`,  
`PURCHASE_EXPORT`, `REQUISITION_CREATE/APPROVE/CONVERT_TO_PO`

## Tables

`purchase_requisitions(_items)`, `purchase_orders(_items)`, `goods_receipts(_items)`,  
`purchase_invoices`, `purchase_returns(_items)`, `purchase_payments`, `vendor_ledgers`,  
`purchase_attachments`, `purchase_notes`, `purchase_status_logs`

---

# MODULE 5 — Inventory Management

**Central operational engine. Centralized inventory engine + distributed branch ownership. Inventory belongs to Branch + Warehouse , never globally.**

## Stock Buckets

Available, Reserved, Damaged, Transit, Blocked .

## Submodules

1. **Inventory Ledger (Most Critical)** — No direct stock editing. Every movement creates a ledger entry with: product, variant, branch, warehouse, movement type, qty change, before/after qty, reference#, user, timestamp.
2. **Branch-wise Inventory** — Separate stock/visibility/reports per branch. Users only access authorized branches.
3. **Warehouse Management** — Types: `MAIN_WAREHOUSE`, `STORE_WAREHOUSE`, `DAMAGE_WAREHOUSE`, `TRANSIT_WAREHOUSE` .
4. **Movement Types** — `PURCHASE`, `SALE`, `RETURN`, `TRANSFER_IN`, `TRANSFER_OUT`, `ADJUSTMENT`, `DAMAGE`, `INTERNAL_CONSUMPTION`, `OPENING_STOCK` (+ future: `MANUFACTURING`, `ASSEMBLY`, `BUNDLE`, `SUBSCRIPTION_RESERVE`).
5. **Stock Transfers** — HO↔Branch, Branch↔Branch, Warehouse↔Warehouse. Statuses: `PENDING` → `IN_TRANSIT` → `RECEIVED` / `CANCELLED` . Source reduces immediately; destination increases only after receiving confirmation.
6. **Reservations** — For online orders, draft invoices, future bookings. Auto-expire on cancel/abandon.
7. **Damaged Inventory** — Reasons: `BROKEN`, `EXPIRED`, `LEAKAGE`, `QUALITY_ISSUE` . Moves to damage warehouse.
8. **Adjustments** — Permission + reason + audit. Reasons: `PHYSICAL_COUNT`, `SYSTEM_CORRECTION`, `MISPLACED`, `INITIAL_SETUP` . Creates ledger entry.
9. **Opening Stock** — Creates ledger entry, separately identifiable.

10. **Batch & Expiry** (future-ready Day 1 schema) — Batch#, mfg date, expiry, FIFO.
11. **Barcode** — Generation, scanning, search, POS integration; future QR + label printing.
12. **Audits** — Stock count sheets, variance reports.
13. **Low Stock Alerts** — Branch-wise thresholds; future email/WhatsApp/push.
14. **POS Visibility** — Current branch only. POS sales consume local branch inventory only. Optional permissioned cross-branch lookup.
15. **Online Visibility (Location-Aware)** — Customer pincode → nearest fulfillment branch → relevant inventory.  
**Never show global stock.** Maintain `online sellable quantity = physical - reserved buffer` .  
 Per-branch `online_fulfillment_enabled = YES/NO` . Fulfillment priority: Nearest Branch → City Warehouse → HO Warehouse.
16. **Reports** — Stock Summary, Ledger, Branch/Warehouse-wise, Low Stock, Damaged, Fast/Slow/Dead Stock, Valuation, Expiry, Online Fulfillment.
17. **Offline Inventory Sync** — Queue-based, conflict-resolving.
18. **Valuation** — FIFO, AVERAGE\_COST (avoid LIFO initially).
19. **Negative Stock Prohibition (Module-level restatement)** — Hard-block at every entry point (POS online + offline, online checkout, transfers, adjustments). Inventory may reach 0, never below. Not a warning — a strict system constraint.

## Permissions

`STOCK_VIEW`, `STOCK_TRANSFER`, `STOCK_ADJUST`, `STOCK_AUDIT`, `STOCK_EXPORT`

## Tables

`inventory`, `inventory_ledger`, `inventory_movements`, `stock_transfers(_items)`, `warehouses`, `warehouse_stocks`, `damaged_inventory`, `inventory_adjustments`, `inventory_reservations`, `batch_inventory`, `opening_stock`, `stock_audits`

---

# MODULE 6 — Online Store

**Digital commerce channel. Same inventory/pricing/customers/orders/wallet/referral/offers as the rest of the platform — ecommerce is a sales channel, not a separate system.**

## Submodules

1. **Customer Storefront** — Homepage, banners, categories, listings, offers, search, featured. Mobile-first.
2. **Product Catalog** — Products/variants/pricing/stock/details/offers; `ONLINE_VISIBLE` flag hides POS-only products.
3. **Intelligent Location Selection** — Auto-suggest from last pincode → saved address → GPS → manual.  
**Customer confirmation mandatory.** Inventory loads only after location confirmation.
4. **Mandatory Pincode-Based Shopping** — Inventory, delivery, offers, fulfillment all pincode-aware.
5. **Location-Aware Inventory Visibility** — Customer pincode → nearest fulfillment branch. Never show global stock.
6. **Search & Filters** — Keyword/category/brand/price/availability; future: AI recommendations, voice.
7. **Shopping Cart** — Validates live inventory before checkout.
8. **Checkout** — Address, delivery validation, payment, summary. Fast & minimal.
9. **Customer Registration & Access** — Only registered customers can place orders. Guests can browse & explore.
10. **Intelligent Registration Approval** — Serviceable pincode → auto-approve after OTP. Non-serviceable → store in CRM + generate Admin Serviceability Request (not rejection).
11. **Pincode Serviceability Engine** — Zones: `STANDARD`, `EXTENDED`, `PREMIUM_SPECIAL`, `NON_SERVICEABLE` .

12. **Delivery Charge Engine** — Per pincode: delivery fee, extra fee, free threshold, ETA, COD availability, assigned branch.
13. **Minimum Order Value (MOV)** — Varies by pincode/zone.
14. **Delivery & Fulfillment** — Customer location → fulfillment branch → reservation → confirm → dispatch.  
Future models: `STANDARD_DELIVERY`, `SAME_DAY_DELIVERY`, `STORE_PICKUP` .
15. **Inventory Reservation** — During order placement; auto-expires.
16. **Order Management** — `PENDING` → `CONFIRMED` → `PACKED` → `SHIPPED` → `DELIVERED` → `COMPLETED` (+ `CANCELLED/RETURNED/REFUNDED`). Audit trail on each event.
17. **Payment Gateway** — UPI, cards, net banking, COD, wallet; future: Razorpay/PhonePe/Paytm. Modes: `ONLINE_PAYMENT`, `COD`, `WALLET` .
18. **Offers & Coupons** — Channel types: `ONLINE`, `POS`, `BOTH` .
19. **Referral Integration** — Reward triggers only after successful completion.
20. **Wallet Integration** — Redemption, referral rewards, future cashback.
21. **Customer Account Panel** — Profile, addresses, orders, wallet, referrals, saved, reorder, invoice download, tracking.
22. **Wishlist** — Future: stock & price-drop alerts.
23. **Notifications** — Order/payment/shipping/delivery. Future: SMS/Email/WhatsApp/Push.
24. **SEO & Marketing** — SEO URLs, meta, sitemap; future: abandoned cart, recommendations, campaigns.
25. **Reports** — Online sales, pincode-wise, branch fulfillment, delivery profitability, cart abandonment, offer usage, location-wise demand.
26. **Security** — Secure checkout, payment validation, fraud prevention, audit logs. Never store sensitive payment data directly.
27. **Customer Registration Approval Flow (origin)** — Statuses originate here and propagate to CRM: `PENDING_SERVICEABILITY` → `ACTIVE` / `BLOCKED` / `INACTIVE` . Non-serviceable pincode → CRM record + admin serviceability request (never auto-rejected).
28. **Abandoned Cart Recovery** (future) — Dedicated submodule: detection, recovery campaigns, recovery analytics, scheduled reminders via SMS/Email/WhatsApp/Push.

## Tables

`online_orders(_items)`, `customer_addresses`, `shopping_carts`, `cart_items`, `wishlists(_items)`, `online_payments`, `coupons`, `coupon_usages`, `serviceable_pincodes`, `online_fulfillment_logs`, `order_status_logs`, `customer_serviceability_requests`, `abandoned_carts`

## MODULE 7 — Retail POS

**Physical retail sales channel. Branch-specific. Uses same inventory/pricing/offers/customer/wallet engines.**

### Submodules

1. **Billing Interface** — Left: search/categories/products; Right: bill/totals/payment. Fast, minimal.
2. **Product Selection** — Barcode scan, manual search (name/SKU/category/partial), category selection, frequent products. **Barcode is optional, never mandatory.** Manual selection fully supported.
3. **Strict Stock-Aware Billing** — Real-time validation. **Negative stock strictly not allowed.** Stock may reach 0 but never below. Hard-block, not warning.
4. **Category-Based Quick Billing** — For touch-screen / fast counters.
5. **Frequent / Recent / Favorite Products** — Improves billing speed.
6. **Variant-Aware Billing** — Size/pack selection per product.
7. **Customer Management** — Existing lookup by mobile, new customer creation, optional tagging. Types: `WALK_IN`, `REGISTERED`, `PARTNER` .



8. **Offers & Discounts** — Channels `ONLINE / POS / BOTH` . Flat, %, bill-level, product-level. Validation includes POS channel.
9. **Payment Collection** — `CASH, UPI, CARD, WALLET, MIXED` . Supports multi-payment billing (e.g. ₹500 Cash + ₹300 UPI).
10. **Wallet Integration** — Redemption, loyalty, referral usage; ledger-driven.
11. **Invoices** — Thermal, A4, digital. Branch + GST + items + payment + savings. Future: WhatsApp/email/QR receipts.
12. **Returns & Exchanges** — Full/partial/exchange. Reasons: `DAMAGED, WRONG_PRODUCT, CUSTOMER_RETURN` . Restores branch inventory.
13. **Offline POS Support (Most Critical)** — Must work without internet. Offline billing → local queue → sync later. Offline POS is branch-scoped; offline inventory remains locally consistent (blocks once local stock = 0). Requires: local inventory + product cache, offline invoice numbering, sync conflict handling.
14. **Inventory Consumption** — Current branch only; never global. Optional permissioned cross-branch lookup.
15. **Cashier & Shift Management** — Login, shift open/close, opening/closing balance, cash mismatch tracking.
16. **Reports** — Daily sales, cashier-wise, branch-wise, payment-mode, returns, discount, item-wise.
17. **Notifications** (future) — Low stock, shift, billing alerts.
18. **Security & Audit** — All billing creates audit trail; cashier tracking, refund approvals.
19. **Performance** — Local caching, lightweight UI, fast search, keyboard shortcuts.
20. **POS Device Registration / Terminal Binding** — POS billing only from registered devices (cross-reference Module 19). Each terminal is mapped to a branch; unregistered devices are hard-blocked from billing.
21. **Tax-Inclusive vs Tax-Exclusive Billing** — Honors product/price-list tax mode from Module 1. POS surfaces correct breakup on bill; configurable globally and per branch.

## Permissions

`POS_BILLING, POS_RETURN, POS_DISCOUNT, POS_CASHIER_CLOSE, POS_REPORT_VIEW, POS_REFUND_APPROVE`

## Tables

`pos_sales(_items), pos_payments, pos_returns(_items), cashier_shifts, cashier_sessions, pos_devices, offline_sync_queue`

---

# MODULE 8 — Sales & Order Management

**Central commerce transaction engine. All channels (Online, POS, future Wholesale, future Direct) share inventory, customers, offers, wallet, referral, reporting.**

## Submodules

1. **Unified Sales Engine** — All sales flow into one central system.
2. **Channels** — `ONLINE, POS, WHOLESALE, MANUAL` . Affects offers, workflows, invoice flow, reporting.
3. **Sales Order Flow** — Created → Inventory Reserved → Payment Validated → Processing → Dispatch → Delivery → Completion.
4. **Status Flow** — `PENDING → CONFIRMED → PROCESSING → PACKED → SHIPPED → DELIVERED → COMPLETED` (+ `CANCELLED/RETURNED/REFUNDED/FAILED`). Configurable.
5. **Inventory Integration** — Online: reserve → confirm → deduct on fulfillment. POS: instant deduction. Overselling hard-blocks order/billing.
6. **Branch-wise Sales** — Every sale tagged with Branch + Warehouse. Branch-aware analytics.
7. **Customer Sales History** — Unified omnichannel profile.
8. **Invoice Management** — GST/retail invoices, branch-wise series, digital + printable. Future: e-invoicing, e-way bill.

9. **Payment Management** — CASH, UPI, CARD, BANK\_TRANSFER, COD, WALLET, MIXED . **Payment methods are branch-configurable.** POS shows only branch-enabled methods. Online checkout loads methods based on fulfillment branch. All payments create payment ledger entries.
10. **Returns & Refunds** — Online + POS + Exchanges. Reasons: DAMAGED, WRONG\_PRODUCT, CUSTOMER\_RETURN, QUALITY\_ISSUE . Restores correct branch inventory.
11. **Exchange Management** — Product/variant replacement with amount adjustment.
12. **Wallet Integration** — Redemption, cashback, referral commission, future loyalty.
13. **Referral & Partner Integration** — Commission only after successful order completion.
14. **Delivery & Fulfillment Tracking** — Branch-aware. Future: delivery partner integration, live tracking, route optimization.
15. **Omnichannel Offers** — Channels: ONLINE, POS, BOTH . Product/cart/referral/branch/festival offers.
16. **Notifications** — Order, payment, dispatch, delivery, refund.
17. **Analytics & Reports** — Sales summary, channel/branch/product/payment-wise, returns/refunds, wallet, referral; omnichannel customer behavior, repeat purchase, LTV.
18. **Offline Sync** — Queue locally, preserve inventory integrity, branch mapping, invoice consistency.
19. **Security & Audit** — All commerce actions create audit trail.
20. **Future Wholesale** — Architecture ready for B2B/dealers/distributors/bulk pricing.

## Permissions

SALES\_VIEW, SALES\_CREATE, SALES\_RETURN, SALES\_REFUND, SALES\_EXPORT, ORDER\_MANAGE, DELIVERY\_MANAGE

## Tables

sales\_orders(\_items), sales\_payments, sales\_returns(\_items), sales\_refunds, sales\_exchanges, sales\_invoices, sales\_channels, sales\_status\_logs, delivery\_tracking, customer\_sales\_history, branch\_payment\_methods, payment\_channel\_configurations, invoice\_sequences

## Branch-wise Invoice Numbering

Each branch maintains its own invoice number series (with configurable prefix, padding, financial-year reset). Schema: invoice\_sequences (branch\_id, channel, prefix, current\_number, fy\_start, fy\_end, reset\_policy) .

---

# MODULE 9 — CRM & Customer Management

**Central customer intelligence. One unified customer system across Online, POS, Referral, Manual, future Wholesale.**

## Submodules

1. **Central Customer Database** — One unified profile linking all sales channels. Mobile = primary unique identifier.
2. **Registration & Onboarding** — Serviceable pincode → auto-approve; non-serviceable → CRM + admin review. Statuses: PENDING\_SERVICEABILITY, ACTIVE, BLOCKED, INACTIVE .
3. **Address Management** — Multiple addresses, history. Pincode drives serviceability/fulfillment/delivery rules. Future: GPS/map.
4. **Customer Segmentation** — RETAIL, PREMIUM, PARTNER, WHOLESALE, FREQUENT\_BUYER . Enables targeted offers, loyalty, analytics.
5. **Referral Relationships** — Successful referrer auto-becomes Partner. Tracks referred customers, hierarchy (non-MLM), earnings, wallet commissions.

6. **Wallet Management** — Types: REFERRAL\_CREDIT, CASHBACK, ORDER\_REDEMPTION, MANUAL\_ADJUSTMENT . Ledger-driven, never directly editable.
7. **Communication History** — SMS, Email, future WhatsApp, support.
8. **Notes & Internal Remarks** — Admin-only.
9. **Purchase History** — Online + POS + Returns + Exchanges + Wallet → 360° view.
10. **Loyalty Readiness** — Future SILVER/GOLD/PLATINUM . Schema future-ready Day 1.
11. **Customer Support** (future) — Tickets, chat, escalations.
12. **Notifications** — Registration, orders, wallet, referrals, campaigns.
13. **Analytics** — Acquisition, repeat rate, top customers, referral performance, LTV, AOV, retention; omnichannel behavior.
14. **Search & Lookup** — Fast by mobile/email/name/referral.
15. **Verification** — Mobile OTP, Email OTP, admin approval (special).
16. **Privacy & Security** — Audit logs, access control, data protection; permission-controlled.
17. **Activity Timeline** — Complete customer history.
18. **Future Wholesale CRM** — Dealers, distributors, B2B accounts, business pricing.

## Permissions

CUSTOMER\_VIEW, CUSTOMER\_CREATE, CUSTOMER\_EDIT, CUSTOMER\_EXPORT, WALLET\_ADJUST, CUSTOMER\_COMMUNICATION

## Tables

customers, customer\_addresses, customer\_segments, customer\_wallets, wallet\_ledger, customer\_referrals, customer\_notes, customer\_communications, customer\_status\_logs, customer\_activity\_logs, customer\_serviceability\_requests

# MODULE 10 — Referral, Partner & Wallet Management

**Growth & incentive engine. Strictly Non-MLM: no binary structure, no levels, no chain income, no forced hierarchy. Any successful referring customer automatically becomes a Partner.**

## Flow

Customer Refers → Referred User Registers → Referred User Purchases → Order Completed Successfully → Commission Credited → Referrer becomes Partner.

## Submodules

1. **Referral Engine** — Sources: CUSTOMER, PARTNER, ADMIN, CAMPAIGN . Methods: code, link, mobile, admin tagging. Each customer gets unique referral code/link. Relationship permanent & auditable.
2. **Automatic Partner Conversion** — Once referral succeeds (referred customer completes order). Statuses: PENDING, ACTIVE, BLOCKED, INACTIVE . No manual activation needed initially.
3. **Partner Profile** — Partner ID, referral code, referred customers, wallet balance, earnings history. Linked to customer profile.
4. **Commission Engine** — Configurable. Types: FIXED, PERCENTAGE . Scope: product, category, order value, campaign. Examples: 5% on Ghee orders; ₹100 on first referral. Rule-driven.
5. **Commission Trigger** — Only after successful order completion (future: after return window). Prevents fake referrals and cancelled-order abuse.
6. **Wallet** — All commissions first credit to wallet. Uses: future payouts, order redemption, cashback, referral earnings. Types: REFERRAL\_COMMISSION, CASHBACK, ORDER\_REDEMPTION, MANUAL\_ADJUSTMENT, PAYOUT\_DEDUCTION . **Ledger-driven; never directly editable.**

7. **Wallet Ledger** — Type, ref ID, amount, before/after balance, timestamp, remarks. Audit trail on every change.
8. **Wallet Redemption** — Online/POS/partial. Configurable: minimum order required, max wallet usage %.
9. **Future Payouts** — Initially commissions stay in wallet. Future: bank/UPI payouts, requests, approval workflows. Statuses: PENDING, APPROVED, PAID, FAILED .
10. **Referral Analytics** — Top referrers, conversion, liabilities, sales, ROI.
11. **Campaign-Based Referrals** — Festive bonuses (e.g. Diwali → Double rewards), category-based.
12. **Validation Rules** — Self-referral blocked, duplicate prevention, fraud detection.
13. **Visibility** — Admin views referrer→referred mapping, performance, liabilities.
14. **Notifications** — Referral success, wallet credit, payouts, campaigns.
15. **Security & Audit** — Commission, wallet, payout, fraud tracking.
16. **Future Loyalty Integration.**

## Permissions

REFERRAL\_VIEW, REFERRAL\_CONFIGURE, WALLET\_VIEW, WALLET\_ADJUST, PAYOUT\_APPROVE, PARTNER\_MANAGE

## Tables

partners, partner\_referrals, referral\_rules, wallets, wallet\_ledger, commission\_transactions, partner\_status\_logs, referral\_campaigns, future\_payout\_requests, referral\_fraud\_logs

# MODULE 11 — Notifications & Communication Management

**Central communication engine. All modules trigger notifications through one unified, event-driven, template-based system.**

## Channels

SMS, EMAIL, WHATSAPP, PUSH\_NOTIFICATION, IN\_APP

## Submodules

1. **Central Engine** — Triggers from order placed, payment success, OTP, wallet credit, referral success, low stock, etc.
2. **OTP Management** — Mobile/Email OTP. Channel configurable from Super Admin. Smart: if both mobile & email exist and admin configured email OTP → use email. Expiry, resend, retry limit, rate limit. Auditable.
3. **SMS** — Transactional + OTP (separate from promotional). Future: campaigns.
4. **Email** — OTP, invoices, orders, wallet, reports. Future: newsletters, journeys.
5. **WhatsApp** (future-ready) — Order updates, OTP, wallet, delivery. Plan for WhatsApp Business API.
6. **Push** (future) — App alerts, offers, reminders.
7. **In-App** — Dashboard alerts (low stock, pending approval, new referral, new order).
8. **Templates** — Types: SMS/Email/WhatsApp/Push. Dynamic vars ( {CUSTOMER\_NAME}, {ORDER\_ID}, {OTP}, {WALLET\_AMOUNT} ). Multi-language ready, versioned, activatable. Admin-editable.
9. **Event-Driven Triggers** — Decoupled from business logic.
10. **Communication Logs** — Recipient, channel, template, status, sent time, delivery status.
11. **Retry & Failure** — Queue-based, retry queue, future provider failover.
12. **Campaign Messaging** (future) — Festive, referral, launches. Opt-in/opt-out required.
13. **Customer Preferences** — Channel-wise consent.
14. **Internal Admin Notifications** — Low stock, refund approvals, non-serviceable requests, payout approvals, failed sync.
15. **Scheduling** (future).

16. **Multi-language** — English, Hindi, Marathi.
17. **Security & Compliance** — Sensitive comms secure & auditable.
18. **Analytics** — Delivery, open, click rate (future), campaign performance, OTP success.

## Permissions

NOTIFICATION\_VIEW, NOTIFICATION\_SEND, TEMPLATE\_MANAGE, CAMPAIGN\_MANAGE, OTP\_CONFIGURATION, COMMUNICATION\_REPORTS

## Tables

notification\_templates, notification\_logs, otp\_logs, communication\_preferences, notification\_queues, campaigns, campaign\_recipients, admin\_notifications

---

# MODULE 12 — Finance, Accounts & Ledger Management

**Financial control & accounting engine. Ledger-driven; every financial transaction creates a ledger impact. Integrates with purchases, inventory, sales, wallet, payments, returns, referrals, expenses.**

## Submodules

1. **Chart of Accounts** — Types: ASSET, LIABILITY, INCOME, EXPENSE, EQUITY . Configurable.
2. **Ledger Management** — Every transaction creates ledger entries (debit/credit accounts, amount, branch, ref, timestamp, narration). Immutable & auditable.
3. **Customer Ledger** — Purchases, refunds, wallet usage; future outstanding for credit/B2B.
4. **Vendor Ledger** — Purchases, payments, outstanding, returns. Integrates with purchase module.
5. **Branch-wise Accounting** — Branch sales/expenses/profitability/collections.
6. **Payment Accounting** — Cash/UPI/Cards/COD/Wallet. Branch-configurable. Every payment hits ledger.
7. **Wallet Accounting** — Wallet is **company liability** until redeemed/paid out. Examples: Referral Credit → Dr Referral Expense / Cr Wallet Liability. Never directly editable.
8. **Expense Management** — Categories: RENT, SALARY, DELIVERY, MARKETING, UTILITY, MISCELLANEOUS . Receipt upload, future approval workflow, branch-tagged.
9. **Tax & GST** — Calculations, breakup, summaries. Future: e-invoicing, GST returns, HSN codes, e-way bills.
10. **Profitability** — Branch/product/category/delivery profitability. Top products, most profitable branch, high expense zones.
11. **Cash & Bank** — Balances, deposits, withdrawals, transfers; future reconciliation & auto statement import.
12. **Refund & Return Accounting** — Affects inventory + sales + accounting + tax.
13. **Referral Accounting** — Commission liabilities, wallet credits, future payout accounting.
14. **Financial Reports** — P&L, Balance Sheet, Cash Flow, Trial Balance, Sales Register, Purchase Register, GST, Expenses, Branch Profitability, Wallet Liability; omnichannel revenue analytics.
15. **Audit & Logs** — Immutable, auditable.
16. **Adjustments** — Correction entries, journal entries, reversals; require permissions, remarks, audit logs.
17. **Future Accounting Integration** — Tally, Zoho Books, ERP exports, CA reporting.
18. **Future Credit System** — Customer/vendor/dealer credit limits; receivable/payable schema ready.
19. **TDS (Tax Deducted at Source)** — Schema-ready from Day 1 (cross-reference Module 3 vendor tax). Vendor-level TDS applicability, section codes, deduction rates, TDS ledger entries, future TDS returns / Form 26Q readiness.
20. **Opening Balance & Year-End Closing** — Opening balance entries per account at FY start; year-end closing flow (P&L → Retained Earnings, ledger lock for closed FY, comparative reports across FYs). Locked FYs are immutable; reopening requires Super Admin + audit log.

## Permissions

FINANCE\_VIEW, LEDGER\_VIEW, EXPENSE\_MANAGE, PAYMENT\_VIEW, PAYMENT\_ADJUST, FINANCIAL\_REPORTS, JOURNAL\_ENTRY

## Tables

chart\_of\_accounts, ledger\_entries, customer\_ledgers, vendor\_ledgers, wallet\_ledgers, branch\_accounts, expenses, expense\_categories, tax\_configurations, financial\_adjustments, bank\_accounts, cash\_accounts, financial\_reports

---

## MODULE 13 — Reports, Analytics & Business Intelligence

**Real-time, actionable, branch-aware. Integrates across sales, inventory, finance, CRM, referrals, fulfillment, procurement, operations.**

### Submodules

1. **Role-based Dashboards** — Super Admin (Global), Branch Manager (Branch Ops), Purchase Team (Procurement), Finance (Financial), POS Cashier (Daily Sales), Partner (Referral Earnings). KPI cards, charts, alerts, pending actions.
2. **Sales Analytics** — Daily/monthly/channel/branch/product; omnichannel revenue analysis.
3. **Inventory Analytics** — Current stock, aging, low stock, valuation, branch summary; smarter procurement planning.
4. **Purchase Analytics** — Vendor-wise, trends, outstanding, returns; procurement efficiency.
5. **Financial Analytics** — P&L, expenses, cash flow, branch profitability, wallet liability.
6. **CRM/Customer Analytics** — Top customers, repeat, retention, AOV; omnichannel behavior.
7. **Referral & Partner Analytics** — Top referrers, conversion, commissions, partner earnings. Non-MLM structured.
8. **Delivery & Fulfillment Analytics** — Pincode-wise, delivery cost, fulfillment efficiency, branch load.
9. **Wallet & Incentive Analytics** — Balance summary, commission liability, usage.
10. **Operational Alerts & Insights** — Low stock, pending approvals, failed sync, unusual activity.
11. **Branch-wise Analytics** — All major analytics remain branch-aware.
12. **Export & Download** — Excel, PDF, CSV with filters (date, branch, category, customer, payment mode, channel).
13. **Scheduled Reports** (future) — Daily/weekly/monthly emailed summaries.
14. **Predictive Analytics** (future) — Demand forecasting, stock prediction, smart reordering, recommendations.
15. **KPI Tracking** — Revenue growth, inventory turnover, AOV, retention.
16. **Audit & Data Integrity** — Stock mismatches, failed syncs, unusual refunds, manual adjustments.
17. **Real-time Readiness** — Live dashboards, POS monitoring.
18. **Mobile Dashboard Readiness.**
19. **Multi-channel Analytics** — Online, POS, future wholesale/distributor.

### Permissions

REPORTS\_VIEW, ANALYTICS\_VIEW, FINANCIAL\_REPORTS, EXPORT\_REPORTS, DASHBOARD\_MANAGE, KPI\_VIEW

## Tables

dashboard\_configurations, analytics\_snapshots, scheduled\_reports, report\_exports, kpi\_definitions, branch\_kpis, analytics\_alerts

---

## MODULE 14 — Delivery, Logistics & Fulfillment Management

**Fulfillment & logistics control engine. Branch-aware + pincode-aware.**

### Submodules

1. **Fulfillment Engine** — Flow: Order → Fulfillment Branch Selection → Reservation → Picking → Packing → Dispatch → Delivery. Branch-scoped.
2. **Intelligent Branch Selection** — Priority: Nearest Branch → Mapped Fulfillment Branch → Warehouse → HO. Per-branch `online_fulfillment_enabled` flag.
3. **Pincode-based Routing** — Zones: `STANDARD`, `EXTENDED`, `PREMIUM_SPECIAL`, `NON_SERVICEABLE` . Pincode controls fulfillment branch, delivery fee, MOV, ETA, payment methods.
4. **Picking** — Picklist gen, inventory validation, status tracking.
5. **Packing** — `PENDING` → `PACKING` → `PACKED` → `FAILED` . Required before dispatch.
6. **Dispatch** — Creation, assignment, scheduling. Branch-linked.
7. **Delivery Status** — `PENDING` → `PACKED` → `DISPATCHED` → `OUT_FOR_DELIVERY` → `DELIVERED` (+ `FAILED/RETURNED/CANCELLED`). Audit trail.
8. **Delivery Charges** — Pincode-based; depends on zone/branch/order value/delivery type.
9. **Delivery Partner Integration** (future) — Dunzo, Porter, Shiprocket, local partners.
10. **Hyperlocal Delivery** — Same-day, local branch dispatch, hyperlocal routing. Types: `STANDARD`, `SAME_DAY`, `EXPRESS`, `STORE_PICKUP` .
11. **Delivery Personnel** (future) — Assignment, tracking, performance, delivery agent app, OTP confirmation, GPS.
12. **Store Pickup / Click & Collect** (future) — Reduces delivery costs.
13. **Failed Delivery Handling** — Reasons: `CUSTOMER_UNAVAILABLE`, `WRONG_ADDRESS`, `PAYMENT_ISSUE` . Trackable, auditable.
14. **Return-to-Origin (RTO)** (future).
15. **Logistics Cost Tracking** — Cost per delivery, pincode profitability, branch logistics expense.
16. **Fulfillment Notifications** — Packing, dispatch, delivery.
17. **Fulfillment Analytics** — Speed, packing efficiency, branch dispatch load, success rate, avg delivery time.
18. **Offline Fulfillment Readiness**.
19. **Security & Audit**.
20. **COD Reconciliation** — Cash collected by delivery agent → branch cash ledger → finance reconciliation. Statuses: `COLLECTED` → `DEPOSITED_TO_BRANCH` → `RECONCILED` / `DISCREPANCY` . Variance triggers audit alert. Branch-scoped, fully traceable.
21. **Delivery Slot Booking** (future) — Customer-selectable delivery slots (e.g. morning/evening/specific windows), per-pincode slot availability, branch capacity planning, slot-based dispatch scheduling.

### Permissions

`FULFILLMENT_VIEW`, `FULFILLMENT_PROCESS`, `DISPATCH_MANAGE`, `DELIVERY_TRACK`, `PICKLIST_MANAGE`, `DELIVERY_REPORTS`

### Tables

`fulfillment_orders`, `picklists`, `packing_logs`, `dispatches`, `delivery_tracking`, `delivery_zones`, `delivery_charges`, `delivery_failures`, `delivery_agents`, `logistics_costs`

---

## MODULE 15 — Returns, Damages & Quality Control Management

**Inventory recovery & quality assurance. Returned/damaged inventory NEVER directly merges into sellable stock without validation.**

## Submodules

1. **Sales Return** — Online, POS, partial, full. Flow: Request → Validation → Approval → Inspection → Inventory Decision → Refund/Replacement. Types: REFUND, REPLACEMENT, EXCHANGE . Branch-aware.
2. **Purchase Return** — Vendor returns, damaged, rejected; linked to original purchase transaction.
3. **Damaged Stock** — Sources: PURCHASE\_DAMAGE, WAREHOUSE\_DAMAGE, DELIVERY\_DAMAGE, CUSTOMER\_RETURN\_DAMAGE . Moves to separate damage inventory; never sellable.
4. **Expiry Management** — Batch & expiry tracking; near-expiry & expired alerts; expired auto-blocks selling.
5. **QC Workflow** — Statuses: PENDING\_QC, PASSED, FAILED, QUARANTINED . Outcomes: RESTOCK, DAMAGE, RETURN\_TO\_VENDOR, DISPOSE . Only QC-approved returns to sellable.
6. **Inventory Quarantine** — Reasons: QC\_PENDING, DAMAGED, EXPIRED, SUSPECTED\_DEFECT . Isolated.
7. **Reverse Logistics** — Customer pickup returns, branch handling, vendor returns; future delivery partner reverse pickup, RTO.
8. **Replacement Workflow** — Linked transaction history.
9. **Inventory Adjustments** — Stock correction, damage, expiry. Permission + remarks + audit. Negative stock strictly prohibited.
10. **Refund Integration** — Methods: ORIGINAL\_PAYMENT, WALLET, STORE\_CREDIT . Traceable & auditable.
11. **Branch-wise Return Handling** — Restores correct branch inventory.
12. **Damage & Loss Analytics** — Damage %, return trends, expiry losses, branch damage rates.
13. **Expiry & Batch Analytics** — Near-expiry, batch performance, expiry losses.
14. **QC Audit & Traceability**.
15. **Disposal Management** — Reasons: EXPIRED, DAMAGED, LEAKAGE, UNSELLABLE . Financially traceable.
16. **Recall Management Readiness** (future) — Product/batch/safety recalls.
17. **Notifications** — Expiry, QC pending, return approval, damage.
18. **Security & Audit**.

## Permissions

RETURN\_APPROVE, QC\_MANAGE, DAMAGE\_MANAGE, EXPIRY\_MANAGE, INVENTORY\_ADJUST, DISPOSAL\_APPROVE

## Tables

sales\_returns, purchase\_returns, damage\_inventory, inventory\_quarantine, batch\_tracking, expiry\_tracking, qc\_inspections, replacement\_orders, inventory\_adjustments, disposal\_logs

---

# MODULE 16 — HR, Staff, Attendance & Payroll Management

**Workforce, attendance & payroll engine. Role-based + branch-aware + attendance-linked + permission-controlled. Payroll is included in Phase 1.**

## Submodules

1. **Employee Management** — Centralized profile (empld, name, mobile, email, branch, designation, dept, reporting manager, joining date, status). Statuses: ACTIVE, INACTIVE, ON\_LEAVE, TERMINATED . Linked to user account.
2. **Branch-wise Staff Allocation** — Employees mapped to branches.
3. **Attendance Management** — Methods: APPLICATION\_PUNCH, BIOMETRIC, GPS\_MOBILE, MANUAL\_ADMIN, HYBRID . Configurable from Super Admin. Branch-wise method allowed (e.g. Pune Biometric / Warehouse App / Delivery GPS). Biometric vendor-independent (ZKTeco, eSSL, API-based).
  - **Login attendance check:** If employee hasn't punched in → system shows popup with [Punch In Now] / [Remind Later] .



4. **Shift Management** — Types: MORNING, EVENING, FULL\_DAY, CUSTOM . POS integration: Punch In → Cashier Shift Open → Billing Enabled. Optionally require attendance before POS billing.
5. **Leave Management** — Types: CASUAL, SICK, PAID, UNPAID . Workflow: Request → Manager Approval → Applied. Permission-controlled.
6. **Department & Role Mapping** — SALES, PURCHASE, WAREHOUSE, FINANCE, OPERATIONS, ADMIN .
7. **Employee Activity Tracking** — Login, approvals, inventory, billing, payroll. Critical actions create audit trail.
8. **Payroll Management (Phase 1)** — Salary structures, attendance-linked, incentives, deductions, processing, approvals, payslip generation. Components: BASIC, HRA, ALLOWANCE, INCENTIVE, BONUS, DEDUCTION . Structures employee-specific.
9. **Attendance-linked Payroll** — Integrates attendance, leave, shifts (e.g. 2 unpaid leaves → deduction).
10. **Payroll Processing** — Flow: Attendance Finalized → Payroll Generated → Review → Approval → Payslip Generated. Statuses: DRAFT, PROCESSING, APPROVED, PAID .
11. **Payslip Generation** — PDF/Print/Email. Employee details, attendance summary, salary breakup, incentives, deductions, net payable. Employee-accessible.
12. **Incentives & Bonus** — Sales, performance, attendance, future referral.
13. **Salary Deductions** — Types: UNPAID\_LEAVE, ADVANCE, PENALTY, OTHER . Remarks + audit.
14. **Payroll Accounting Integration** — Salary Expense Dr / Salary Payable Cr; on payment: Salary Payable Dr / Bank Cr. Auto-entries.
15. **Employee Dashboard** — Attendance, salary history, payslips, deductions; future reimbursements, tax declarations.
16. **HR Reports** — Attendance, payroll summary, leave, branch staff, shift performance; trends, overtime, productivity, payroll expenses.
17. **Security & Audit** — Salary & HR data highly permission-controlled.
18. **Future Readiness** — PF, ESIC, TDS, GPS attendance, field staff, compliance.

## Permissions

EMPLOYEE\_VIEW/MANAGE, ATTENDANCE\_MANAGE, SHIFT\_MANAGE, LEAVE\_APPROVE, PAYROLL\_VIEW/PROCESS/APPROVE, PAYSリップ\_GENERATE, HR\_REPORTS

## Additional Workforce Features

- **Probation & Notice Period** — Per-employee probation window (configurable), confirmation workflow, notice-period tracking on resignation/termination.
- **Full & Final (F&F) Settlement** — Triggered on TERMINATED / resignation. Calculates: pending salary, leave encashment, deductions, recoveries, bonus, gratuity-readiness. Status flow: INITIATED → CALCULATED → APPROVED → PAID . Permission + audit controlled.
- **Holiday Calendar (per-branch)** — National + regional + branch-specific holidays. Drives attendance, payroll calculation, leave eligibility. Editable yearly; locked once a payroll cycle consumes it.

## Tables

employees, employee\_branches, attendance\_logs, attendance\_configurations, attendance\_devices, biometric\_sync\_logs, shift\_assignments, leave\_requests, salary\_structures, payroll\_cycles, employee\_payroll, payroll\_items, payslips, employee\_activity\_logs, holiday\_calendar, employee\_probation, ff\_settlements

# MODULE 17 — Document, Media & File Management

Central digital asset & document engine. All files centralized, secure, linked to entities/transactions, auditable.

## Submodules

1. **Central Storage** — Types: `IMAGE`, `PDF`, `EXCEL`, `CSV`, `DOCUMENT`, `ZIP` . Files never orphaned; always linked (Product→Images, Invoice→PDF, Employee→Documents, Vendor→Agreements).
2. **Product Media** — Multiple images, variant-wise, sorting, default. Types: `THUMBNAIL`, `GALLERY`, `BANNER`, `VARIANT_IMAGE` . Web & mobile optimized.
3. **Invoice & Document Storage** — Sales/purchase invoices, return docs, payslips, reports. Permanently linked to transactions.
4. **Employee Documents** — ID proofs, agreements, joining docs, certifications. Access-controlled. Future: expiry/renewal alerts.
5. **Vendor & Customer Attachments** — Agreements, GST, KYC, contracts; entity-linked, permission-controlled.
6. **Report Export** — PDF, Excel, CSV; downloadable & traceable.
7. **Media Optimization** — Auto compression, thumbnails, responsive.
8. **File Access Control** — Role/permission/ownership/branch-based (HR docs → HR/Admin only).
9. **File Versioning** (future) — History, rollback (e.g. Agreement v1 / v2).
10. **File Audit & Tracking** — Uploaded/modified by, download/access history.
11. **Storage Structure** — Module-organized folders ( `products/`, `employees/`, `vendors/`, `sales/`, `reports/`, `payroll/` ).
12. **Cloud Readiness** — AWS S3, GCS, Azure, local. Storage-provider-independent.
13. **Media Security** — Secure URLs, protected downloads, future signed URLs.
14. **Bulk Upload** — Product images, document imports, Excel imports; future drag-and-drop, zip extraction.
15. **Generated File Automation** — Auto invoices/payslips/exports, auto-linked to source.
16. **Media Analytics** (future) — Storage usage, upload activity, downloads.
17. **Backup & Recovery** — Automated backups, disaster recovery, redundancy.
18. **OCR & AI Readiness** (future) — Invoice OCR, auto GST extraction, AI classification.
19. **Notifications** — Expiry, upload failure, storage warnings.
20. **Security & Compliance.**

## Permissions

`FILE_UPLOAD`, `FILE_DOWNLOAD`, `FILE_DELETE`, `MEDIA_MANAGE`, `DOCUMENT_VIEW`, `REPORT_EXPORT`

## Tables

`files`, `file_categories`, `file_versions`, `file_access_logs`, `media_assets`, `document_links`, `generated_reports`, `storage_configurations`

---

# MODULE 18 — System Settings, Configuration & Automation Management

**Central control engine. Configurable, not hardcoded. Global defaults + branch overrides + role-based control.**

## Submodules

1. **Global Settings** — Company, currency, timezone, tax, invoice prefix, default language.
2. **Branch-wise Configuration** — Payment methods, fulfillment enablement, attendance methods, pricing policies, POS config, invoice series. Independent per branch.
3. **Feature Toggles** — Wallet, Referral, Online Ordering, Payroll enable/disable; beta features.
4. **Payment Configuration** — Branch-configurable methods; visibility depends on branch + channel + config.
5. **Attendance Configuration** — App/Biometric/GPS/Hybrid; configurable from Super Admin; may vary by branch.

6. **Automation Rules Engine** — Low stock alerts, auto wallet credit, order status notifications, attendance reminders. Rule-driven.
7. **Scheduler & Cron** — Night inventory sync, daily sales reports, payroll reminders. Centrally managed.
8. **Approval Workflow Configuration** — Leave, payroll, refunds (>₹5000 → Manager Approval), inventory adjustments.
9. **Notification Configuration** — Providers (SMS/Email/WhatsApp), OTP expiry, retries.
10. **Inventory Configuration** — Negative stock permanently disabled; reorder alerts, reservation, branch transfer policies.
11. **POS Configuration** — Barcode enablement (optional), offline mode, invoice printing, cashier restrictions. May be branch-specific.
12. **Online Store Configuration** — Delivery rules, pincode settings, MOV (e.g. Extended zone ₹1500), serviceability rules.
13. **Referral & Wallet Configuration** — Commission rules, wallet limits, payout policies, campaigns. Non-MLM.
14. **Finance & Accounting Configuration** — Ledger, tax, invoice sequences, branch rules.
15. **Security & Session Configuration** — Password policies, session timeout (e.g. 30 min), login & device restrictions.
16. **Audit & System Logs** — Critical config changes audit-tracked.
17. **Multi-language & Localization** — English, Hindi, Marathi.
18. **Backup & Recovery Configuration** — Automated backups & schedules.
19. **API & Integration Configuration** — Payment gateways, delivery APIs, biometric APIs, accounting integrations.
20. **Future AI & Smart Automation** — AI alerts, predictive automation, smart reorder, AI analytics.
21. **Invoice Prefix & Financial-Year Reset** — Per-branch invoice prefix, padding, sequence start, automatic FY reset (configurable FY start month), separate series per channel (POS/Online/Wholesale).
22. **Print Template Configuration** — Thermal (58mm/80mm), A4, digital. Header/footer, logo, terms, GST block, QR/UPI block; branch-overrideable. WYSIWYG template editor (future).

## Permissions

SYSTEM\_SETTINGS\_MANAGE, AUTOMATION\_MANAGE, BRANCH\_CONFIGURATION, PAYMENT\_CONFIGURATION, SECURITY\_CONFIGURATION, FEATURE\_TOGGLE\_MANAGE

## Tables

system\_settings, branch\_configurations, feature\_toggles, automation\_rules, scheduler\_jobs, approval\_workflows, payment\_configurations, attendance\_configurations, notification\_configurations, security\_configurations, audit\_logs

---

# MODULE 19 — Super Admin, Security & Audit Control Management

**Enterprise governance & security engine. Every critical action is permission-controlled + auditable + traceable.**

## Submodules

1. **Super Admin Panel** — Centralized enterprise control: global config, branch management, feature toggles, security, audit, operational overrides. All Super Admin actions fully auditable.
2. **RBAC** — Permission-based, not just role-name (e.g. INVENTORY\_VIEW, PAYROLL\_APPROVE, REFUND\_APPROVE, SYSTEM\_SETTINGS\_MANAGE ). Granular & scalable.
3. **Permission Audit** — Tracks permission changes, role assignments, access escalations.
4. **Audit Logging** — Immutable logs for inventory adjustments, refunds, payroll approvals, wallet adjustments, system settings, logins. Fields: user, action, timestamp, IP, branch, device, remarks. Tamper-resistant.

5. **Login & Session Security** — Timeout (e.g. 30 min), device tracking, concurrent session control (e.g. max 1), suspicious login detection.
6. **Device Management** — Authorized devices, branch device mapping, POS terminal restrictions (e.g. POS billing only from registered devices).
7. **Branch-wise Security** — Branch-specific access, restrictions, device policies. Data visibility branch-aware.
8. **Financial Security** — Refund approval hierarchy (e.g. >₹5000 → Manager mandatory), wallet adjustment restrictions, payroll approval controls.
9. **Inventory Security** — Stock adjustment approvals, transfer approvals, audit monitoring.
10. **HR & Payroll Security** — Salary access control, payroll approval restrictions, attendance audit.
11. **POS Security** — Cashier shift validation, refund restrictions, offline sync audit; operator-traceable.
12. **Fraud Prevention** — Suspicious activity alerts, unusual refund monitoring, duplicate transaction detection.
13. **OTP & Verification Security** — Retry limits (e.g. max 5), brute-force prevention, throttling.
14. **API & Integration Security** — Auth, token management, webhook validation.
15. **Data Access Governance** — Restricted exports, sensitive data masking, download permissions.
16. **Operational Monitoring Dashboard** — Active sessions, failed logins, audit alerts, system health.
17. **Backup & Disaster Recovery Governance.**
18. **Compliance Readiness** — GST, HR, financial audit-ready.
19. **MFA Readiness** (future) — OTP MFA, authenticator apps, biometric, trusted device verification.
20. **Security Analytics** — Failed logins, suspicious access, approval anomalies, incidents.

## Permissions

SUPER\_ADMIN, AUDIT\_VIEW, SECURITY\_MANAGE, SESSION\_MONITOR, PERMISSION\_MANAGE, DEVICE\_MANAGE

## Tables

audit\_logs, user\_sessions, device\_registrations, permission\_audits, security\_events, failed\_login\_attempts, otp\_security\_logs, system\_health\_logs, approval\_audit\_logs

# MODULE 20 — Mobile App, Offline Sync & API Integration Management

**Digital connectivity & mobility engine. Mobile-ready + offline-capable + API-driven. System must work during unstable internet, branch connectivity issues, mobile interruptions.**

## Submodules

1. **Mobile App Architecture** — Multi-app ecosystem: Customer App, POS App, future Delivery/Partner/Admin apps. API-first.
2. **Offline-first Architecture** — Critical operations support offline: POS Billing, Attendance Punch, Inventory Lookup, Customer Lookup.
3. **Central Sync Engine** — Offline Action → Local Queue → Internet → Background Sync. Conflict-aware.
4. **Offline Inventory Consistency** — Locally controlled to prevent overselling (Local Stock 5 + Offline Sale 5 → Next billing blocked). Negative stock strictly prohibited.
5. **Offline POS Support** — Local invoice generation, sync queue, local product/inventory/customer caches.
6. **Offline Attendance** — Local punch, sync later.
7. **API Gateway** — Central API layer for web/mobile/POS/third-party.
8. **Auth & API Security** — Token auth, session validation, device authorization, rate limiting. Future: OAuth, API keys, webhook security.
9. **Third-party Integrations** — Razorpay, Shiprocket, ZKTeco, Tally, payment gateways, delivery partners, biometric, accounting, SMS/email providers. Modular.
10. **Push Notifications** — Order delivered, wallet credited, attendance reminders.

- 11. **Device Management** — Registration, trusted devices, branch mapping (POS device → branch linked). Critical ops device-restricted.
- 12. **Local Caching** — Product, inventory, customer, settings caches.
- 13. **Sync Conflict Resolution** — Timestamp + transaction priority logic.
- 14. **Background Sync Processing** — Auto sync, retry queue, failed sync recovery. Resilient & recoverable.
- 15. **API Monitoring** — Response time, sync failure rate, offline queue size.
- 16. **Mobile Performance Optimization** — Lightweight APIs, optimized media, cached requests. Low-bandwidth optimized.
- 17. **PWA Readiness** (future) — Installable web app, offline browser, push.
- 18. **IoT & Smart Device Readiness** (future) — Smart weighing scales, barcode scanners, IoT sensors.
- 19. **Backup & Recovery for Offline Devices** — Local backup, recovery sync, queue restoration.
- 20. **Security & Audit.**

Permissions

API\_MANAGE, DEVICE\_MANAGE, SYNC\_MONITOR, OFFLINE\_OVERRIDE, MOBILE\_APP\_MANAGE

Tables

api\_tokens, device\_registrations, offline\_sync\_queue, sync\_logs, mobile\_sessions, api\_request\_logs, integration\_configurations, offline\_cache\_metadata

Appendix A — Consolidated Status Flows

| Domain                | Flow                                                                                                               |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Purchase Requisition  | DRAFT → SUBMITTED → APPROVED → PARTIALLY_PROCESSED → COMPLETED / REJECTED                                          |
| Purchase Order        | DRAFT → PENDING_APPROVAL → APPROVED → PARTIALLY_RECEIVED → COMPLETED / CANCELLED                                   |
| Stock Transfer        | PENDING → IN_TRANSIT → RECEIVED / CANCELLED                                                                        |
| Online Order          | PENDING → CONFIRMED → PACKED → SHIPPED → DELIVERED → COMPLETED (+ CANCELLED/RETURNED/REFUNDED)                     |
| Sales Order (unified) | PENDING → CONFIRMED → PROCESSING → PACKED → SHIPPED → DELIVERED → COMPLETED (+ CANCELLED/RETURNED/REFUNDED/FAILED) |
| Delivery              | PENDING → PACKED → DISPATCHED → OUT_FOR_DELIVERY → DELIVERED (+ FAILED/RETURNED/CANCELLED)                         |
| Packing               | PENDING → PACKING → PACKED / FAILED                                                                                |
| QC                    | PENDING_QC → PASSED / FAILED / QUARANTINED                                                                         |
| Payroll               | DRAFT → PROCESSING → APPROVED → PAID                                                                               |
| Future Payout         | PENDING → APPROVED → PAID / FAILED                                                                                 |
| Customer (CRM)        | PENDING_SERVICEABILITY → ACTIVE / BLOCKED / INACTIVE                                                               |
| Vendor                | ACTIVE / INACTIVE / BLACKLISTED / ON_HOLD                                                                          |
| Partner               | PENDING → ACTIVE / BLOCKED / INACTIVE                                                                              |

|          |                                           |
|----------|-------------------------------------------|
| Employee | ACTIVE / INACTIVE / ON_LEAVE / TERMINATED |
|----------|-------------------------------------------|

---

## Appendix B — Inventory Movement Types

PURCHASE, SALE, RETURN, TRANSFER\_IN, TRANSFER\_OUT, ADJUSTMENT, DAMAGE, INTERNAL\_CONSUMPTION, OPENING\_STOCK Future: MANUFACTURING, ASSEMBLY, BUNDLE, SUBSCRIPTION\_RESERVE

---

## Appendix C — Channels & Offer Scope

- **Sales Channels:** ONLINE, POS, WHOLESALE (future), MANUAL
  - **Offer Channels:** ONLINE, POS, BOTH
  - **Payment Modes:** CASH, UPI, CARD, BANK\_TRANSFER, COD, WALLET, MIXED
  - **Communication Channels:** SMS, EMAIL, WHATSAPP, PUSH\_NOTIFICATION, IN\_APP
  - **Attendance Methods:** APPLICATION\_PUNCH, BIOMETRIC, GPS\_MOBILE, MANUAL\_ADMIN, HYBRID
  - **Delivery Zones:** STANDARD, EXTENDED, PREMIUM\_SPECIAL, NON\_SERVICEABLE
- 

## Appendix D — Phase 1 Inclusions (Explicit per SRS)

- Payroll is in Phase 1 (Module 16).
  - Wallet is included; future payouts deferred (Module 10).
  - Batch & Expiry: schema future-ready Day 1 (Module 5/15).
  - Loyalty: schema future-ready Day 1 (Module 9).
  - Branch-wise pricing: schema future-ready (Module 1).
  - Multi-language: templates localization-ready Day 1 (Module 11).
  - Cloud storage independence: Day 1 (Module 17).
  - Finance-ready vendor ledger / receivable-payable schema: Day 1 (Modules 3, 12).
  - Offline POS & attendance: Day 1 (Modules 7, 16, 20).
- 

## Appendix E — Cross-cutting Non-Functional & Governance Requirements

These span all modules and are not owned by any single module.

### Data Retention & Archival

- Audit logs: retain **minimum 7 years** (GST/finance compliance), append-only.
- Operational logs (API, sync, OTP): retain **12 months**, then archive.
- Soft-deleted master records: retained indefinitely; never purged.
- Archival store separate from operational DB (cold storage / S3 Glacier-class).

### Multi-currency

- **Non-goal for Phase 1.** System operates in **INR only**. Schema includes a `currency_code` column on monetary tables for future readiness, defaulted to `INR`.

### Localization (UI)

- Notification templates: multi-language Day 1 (English, Hindi, Marathi).
- **UI i18n:** English only in Phase 1; schema/keys structured for future i18n (no hardcoded user-facing strings in code).

### Rate Limiting & API Throttling

- OTP: max 5 attempts / 15 min per identifier; resend cooldown  $\geq$  30s.
- Login: max 10 failed attempts  $\rightarrow$  temporary lock (15 min) + alert.
- Public API: token-bucket per API key; configurable per integration.
- Webhook receivers: signature + replay-window validation.

### Error Catalog & User-Facing Error Codes

- Standardized error code namespace (e.g. `INV-001` inventory, `PAY-001` payment, `AUTH-001` auth).
- Every error: `code`, `http_status`, `user_message`, `internal_message`, `log_ref` .
- Centralized error reference doc maintained alongside this spec.

### Environment & Deployment Topology

- Environments: `dev` , `staging` , `prod` — isolated DBs, secrets, integrations.
- **Single-tenant** architecture (one AsliChoice instance); multi-tenant **not in scope**.
- Branches are logical entities within the single tenant, not separate tenants.

### SLA & Performance Targets

- POS billing add-to-cart / save: < **2s** at p95.
- Online product listing (post-pincode confirm): < **3s** at p95.
- Dashboard / reports first paint: < **5s** at p95.
- OTP delivery: < **30s** end-to-end at p95.
- Sync queue drain (after reconnect): < **5 min** for typical offline POS shift.

### Disaster Recovery — RPO / RTO

- **RPO (Recovery Point Objective):**  $\leq$  **15 min** for transactional data.
- **RTO (Recovery Time Objective):**  $\leq$  **4 hours** for full restoration.
- Daily full backups + continuous transaction-log shipping.
- Quarterly DR drill mandatory; results logged.

### Data Privacy & Right to Be Forgotten

- Soft-delete is the **operational** norm; never deletes financial/audit records.
- **Legal data-erasure path** for customer PII: pseudonymize PII (name, email, mobile, address) on verified request while preserving transactional integrity and tax-mandated records. Erasure action is itself audit-logged.
- Access to PII export tools is permission-gated and audit-logged.

---

## Appendix F — Glossary / Acronyms

| Term | Meaning                           |
|------|-----------------------------------|
| AOV  | Average Order Value               |
| API  | Application Programming Interface |
| B2B  | Business-to-Business              |
| BI   | Business Intelligence             |
| COD  | Cash on Delivery                  |
| CRM  | Customer Relationship Management  |
| DR   | Disaster Recovery                 |

|      |                                                        |
|------|--------------------------------------------------------|
| ERD  | Entity-Relationship Diagram                            |
| ESIC | Employees' State Insurance Corporation                 |
| F&F  | Full & Final Settlement                                |
| FIFO | First-In-First-Out                                     |
| FY   | Financial Year                                         |
| GDPR | General Data Protection Regulation (privacy reference) |
| GPS  | Global Positioning System                              |
| GRN  | Goods Receipt Note                                     |
| GST  | Goods & Services Tax                                   |
| HO   | Head Office                                            |
| HSN  | Harmonized System of Nomenclature (tax code)           |
| IoT  | Internet of Things                                     |
| JWT  | JSON Web Token                                         |
| KPI  | Key Performance Indicator                              |
| LTV  | Lifetime Value                                         |
| MFA  | Multi-Factor Authentication                            |
| MLM  | Multi-Level Marketing                                  |
| MOQ  | Minimum Order Quantity                                 |
| MOV  | Minimum Order Value                                    |
| OCR  | Optical Character Recognition                          |
| OTP  | One-Time Password                                      |
| P&L  | Profit & Loss                                          |
| PAN  | Permanent Account Number                               |
| PF   | Provident Fund                                         |
| PII  | Personally Identifiable Information                    |
| PO   | Purchase Order                                         |
| POS  | Point of Sale                                          |
| PWA  | Progressive Web App                                    |
| QC   | Quality Control                                        |
| RBAC | Role-Based Access Control                              |



|     |                                                                                        |
|-----|----------------------------------------------------------------------------------------|
| RPO | Recovery Point Objective                                                               |
| RTO | Return-to-Origin (logistics) / Recovery Time Objective (DR) — disambiguated by context |
| SKU | Stock Keeping Unit                                                                     |
| SLA | Service Level Agreement                                                                |
| SRS | Software Requirement Specification                                                     |
| TDS | Tax Deducted at Source                                                                 |
| UPI | Unified Payments Interface                                                             |

## Appendix G — Related Documents

- Architecture (legacy / current draft): [doc/old.architecture.md](#)
- Source SRS: [doc/SOFTWARE\\_REQUIREMENT\\_SPECIFICATION\\_ASLI\\_CHOICE.md](#)

*Note: Online Order flow in Appendix A is a **specialization** of the unified Sales Order flow — they are intentionally aligned, not duplicated.*

## Appendix H — Architectural Decisions to Carry into Implementation

1. Use a **single user table** with multi-role mapping.
2. Use a **ledger table per domain** (inventory, wallet, vendor, customer, accounting) — never mutate balances directly.
3. **Inventory updates only on GRN** for purchases; only on confirmed receipt for transfers.
4. **POS: hard-block** when local stock would go negative — including offline mode.
5. **Online inventory loads only after customer confirms location** (pincode).
6. **Branch-aware everything**: payments, fulfillment, pricing, reports, security, attendance, payroll.
7. **OTP destination is computed by the system**, not by the user input field.
8. **Soft delete** for all master records.
9. **Audit log table** with tamper-resistant entries for all critical actions.
10. **API-first** so mobile apps, POS, and integrations all consume the same contracts.